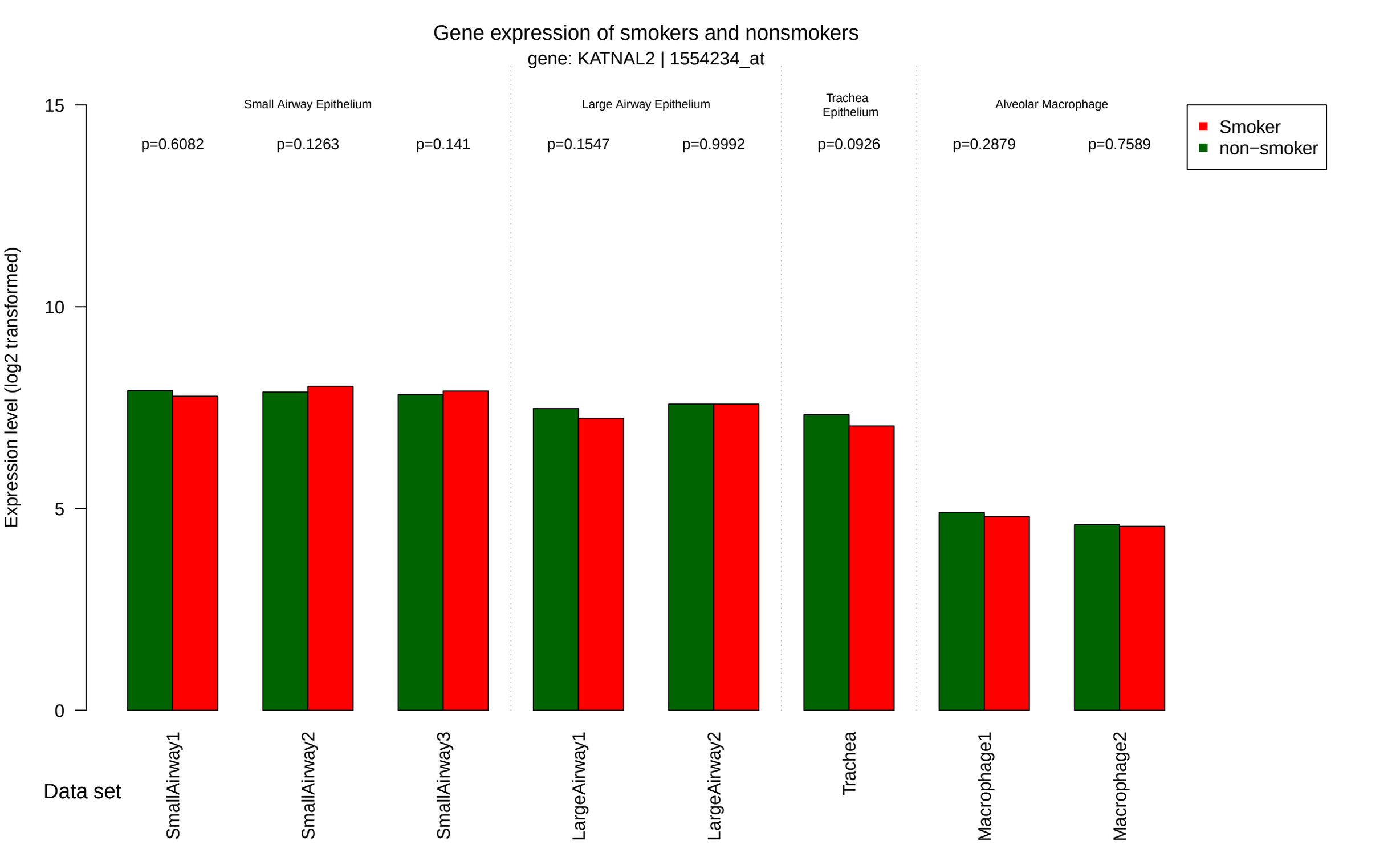
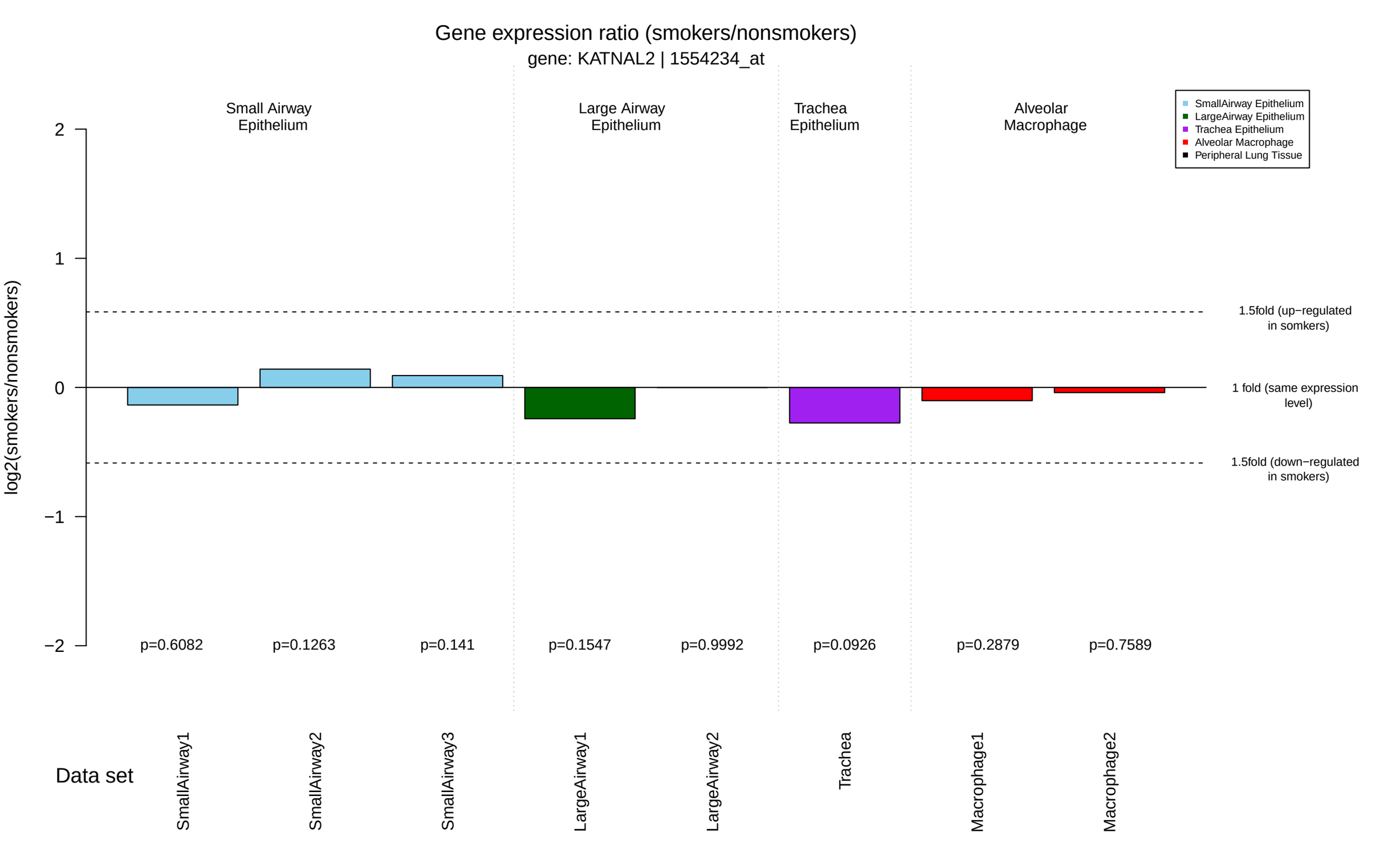
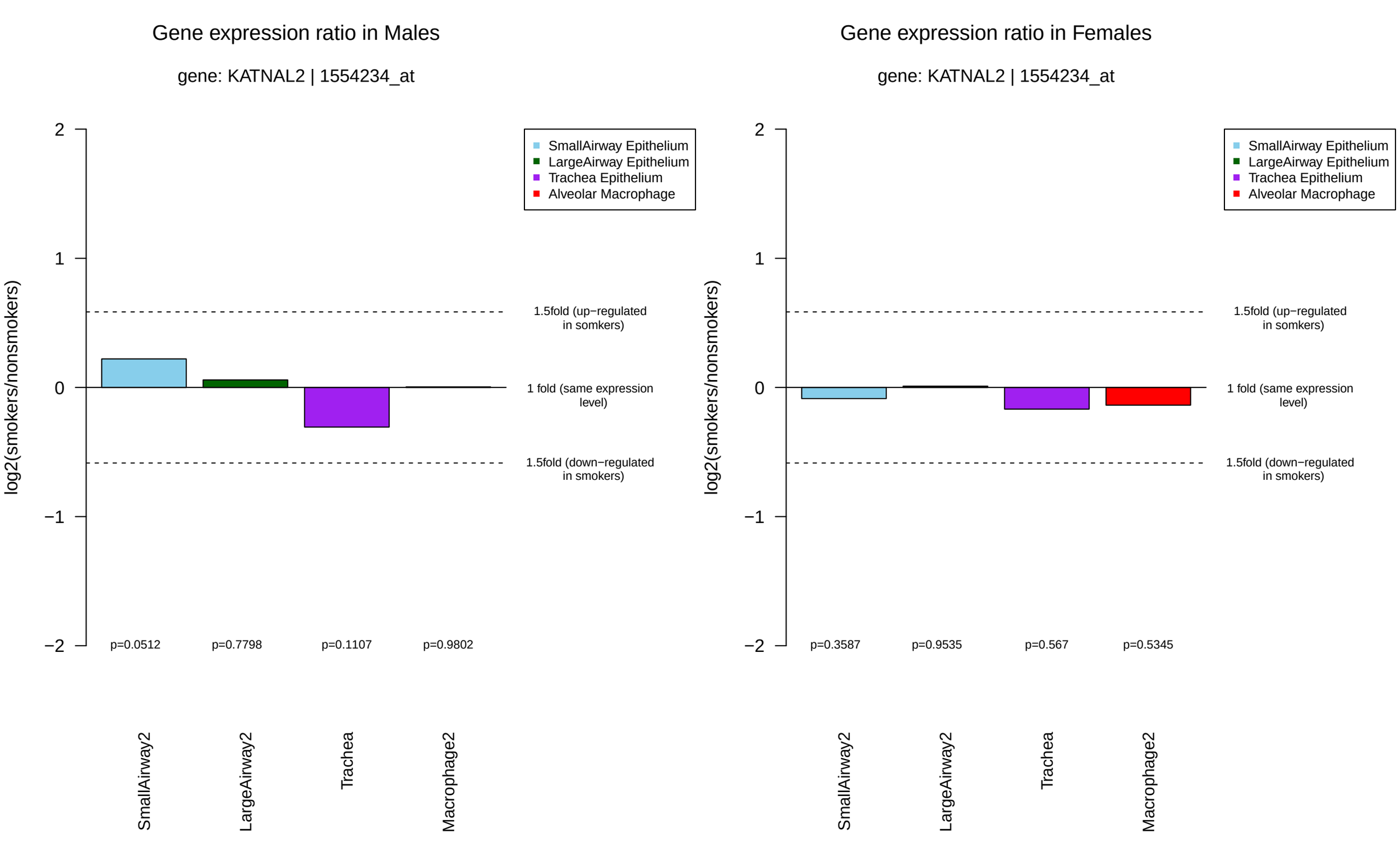
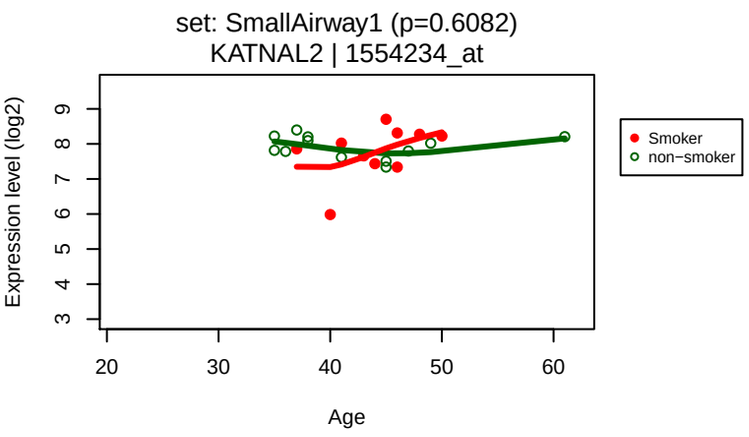
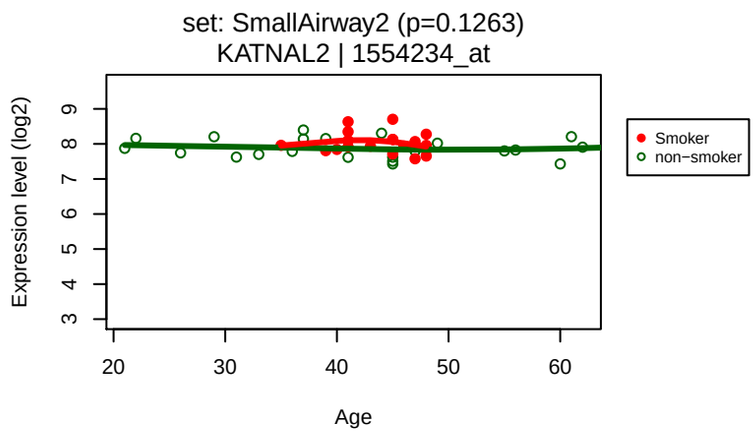
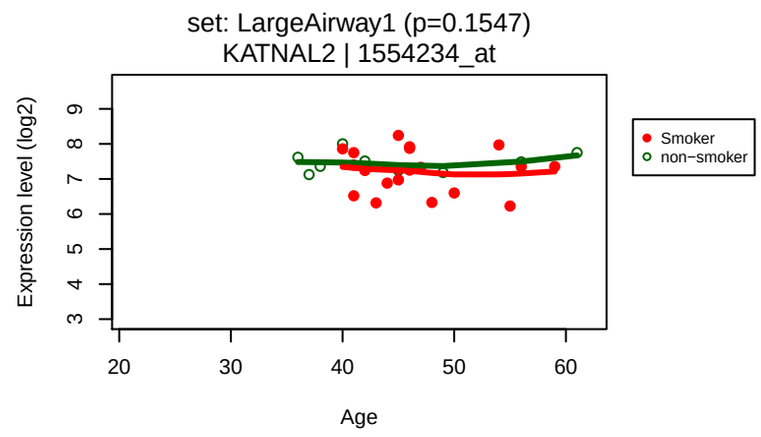

Gene expression of smokers and nonsmokers
gene: KATNAL2 | 1554234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6082
p=0.1263
p=0.141
p=0.1547
p=0.9992
p=0.0926
p=0.2879
p=0.7589
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

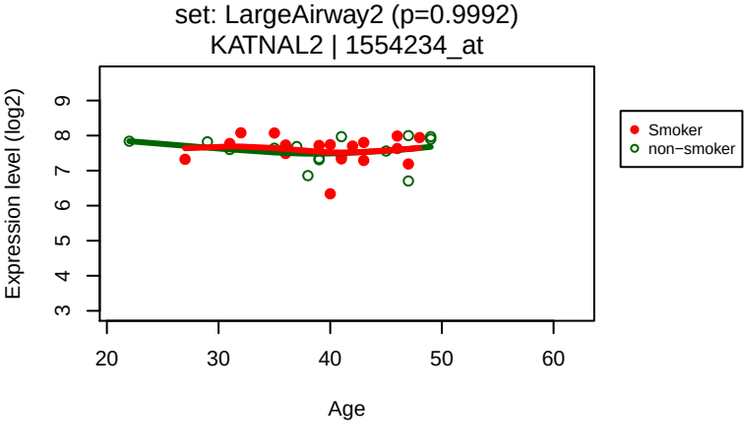

Gene expression ratio (smokers/nonsmokers)
gene: KATNAL2 | 1554234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6082
p=0.1263
p=0.141
p=0.1547
p=0.9992
p=0.0926
p=0.2879
p=0.7589
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

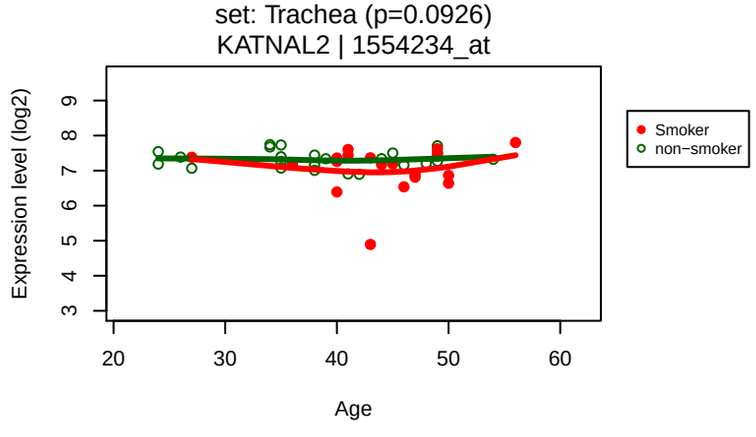

Gene expression ratio in Males
gene: KATNAL2 | 1554234_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0512
p=0.7798
p=0.1107
p=0.9802
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KATNAL2 | 1554234_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3587
p=0.9535
p=0.567
p=0.5345
SmallAirway2
LargeAirway2
Trachea
Macrophage2

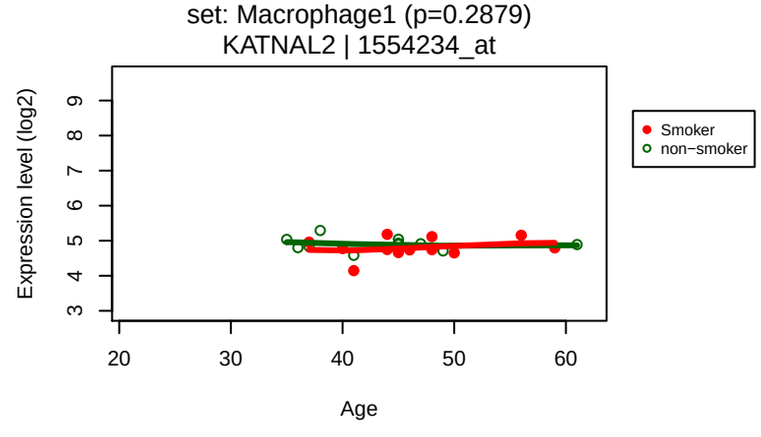

set: SmallAirway1 (p=0.6082)
KATNAL2 | 1554234_at
Expression level (log2)
Age
Smoker
non−smoker

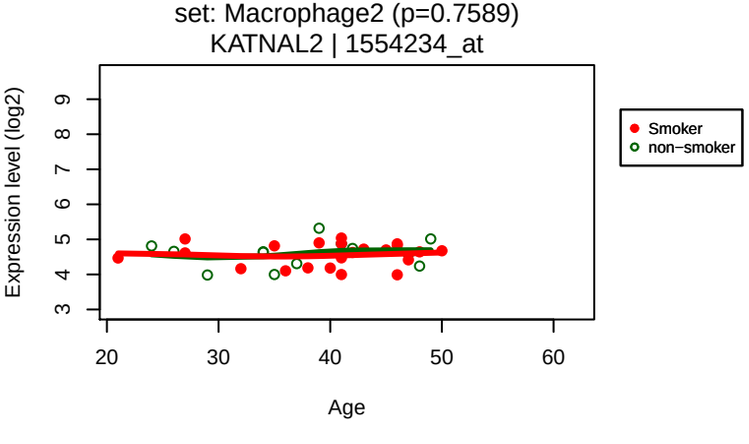

set: SmallAirway2 (p=0.1263)
KATNAL2 | 1554234_at
Expression level (log2)
Age
Smoker
non−smoker

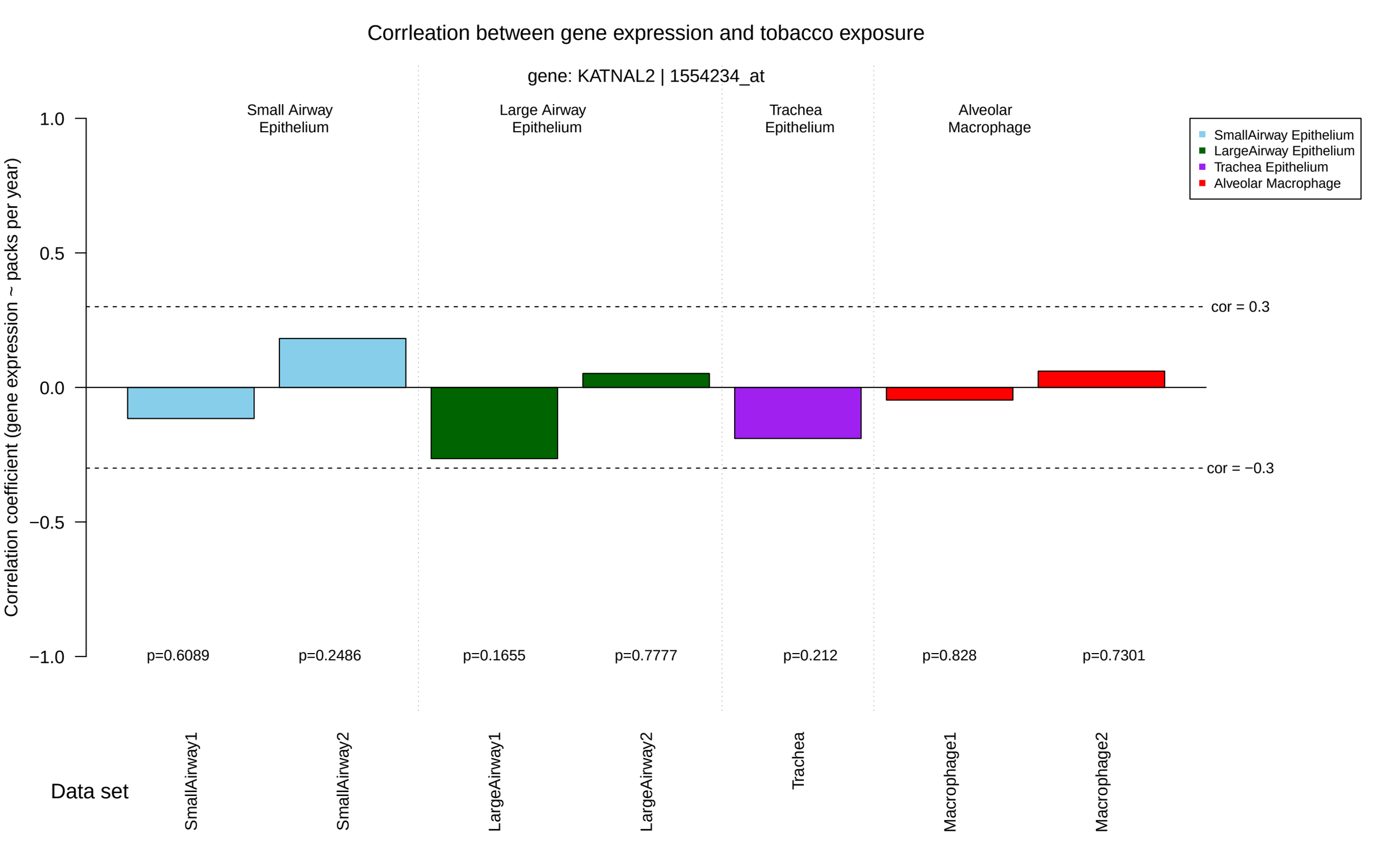

set: LargeAirway1 (p=0.1547)
KATNAL2 | 1554234_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9992)
KATNAL2 | 1554234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0926)
KATNAL2 | 1554234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2879)
KATNAL2 | 1554234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7589)
KATNAL2 | 1554234_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KATNAL2 | 1554234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6089
p=0.2486
p=0.1655
p=0.7777
p=0.212
p=0.828
p=0.7301
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2